Audiophile Sprint Reports

Team:

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Sprint 1

1. **What functionality does the system have at the end of this individual sprint? (2 pts)**

At the end of Sprint 1, the system supported the core audio input and preprocessing features. Users can select an instrument, record audio, stop recording, and save or upload an audio file. The system also performs basic audio processing such as reducing background noise, converting audio formats, and normalizing sound levels. These features allow users to begin recording for future pitch analysis in later sprints.

2. **List the user stories that you successfully implemented during this sprint (1 pt)**

Story ID	User Story
1	Select Instrument
2	Start Recording Audio
3	Stop Recording Audio
4	Save Recorded Audio
5	Upload Audio File
6	Reduce Background Noise
7	Convert Audio Format
8	Normalize Sound Levels

3. **Did you end up making any changes to any of these user stories? Did you break down any of the user stories? Did you identify any new user stories during this sprint and, if so, did you add them to the product backlog or decide to implement them right away? Explain (2 pts)**

During the creation of this Sprint, some user stories required small adjustments, Audio processing such as noise reduction and format conversion were more complex than expected, so

they were broken into smaller implementations. No new user stories were added, but improvements were made to improve clarity and ensure each feature worked smoothly together. All changes were added to the backlog as refinements rather than creating an entirely new story.

4. **What are the "lessons learned" at the end of this sprint? What would you do differently next time? Explain (2 pt)**

We learned a few things during this first sprint. One of which was how to work together as a team and keep developing our functionality forward with the intent of planning for future sprints. Pacing ourselves was another lesson. It could be quite easy to move forward with the next sprint functionalities immediately, but we were able to pace ourselves and complete this sprint with decent space to expand with the next two sprints.

5. **Provide an updated numbered list of all user stories yet to be implemented; indicate pre- and post- conditions (1 pt)**

Story ID	User Story	Pre-Conditions	Post-Conditions
9	Detect Pitch Frequency	Processed audio is available and instrument settings are loaded	The fundamental frequency (Hz) is detected and stored.
10	Identify Letter Notation	Pitch frequency is available	Notes are identified (A, B, C)
11	Evaluate Pitch Accuracy	Notes are identified	Accuracy of pitch is calculated
12	Validate Performance	Accuracy is calculated	Performance is validated
13	Send Data to Groq API	Performance data exists	Data is sent to API
14	Generate Feedback Message	Data is processed by API	Feedback is generated
15	Display Generated Feedback	Feedback exists	Feedback is displayed to users
16	View Feedback Results Page	Feedback is displayed	User can view result page

6. Given the current functionality of the system and taking into account the pre- and post-conditions, identify a subset of user stories to be implemented during the next sprint. Be sure that the cumulative size of the selected user stories is about 1/3 of the size of the full backlog. **Describe the functionality that your (partially implemented) system will be at the end of this sprint.**

Story ID	User Story
9	Detect Pitch Frequency
10	Identify Letter Notation
11	Evaluate Pitch Accuracy
12	Validate Performance

By the end of the next sprint, we will have implemented four more user stories. These will dictate more of what the system itself will do in the backend. Based off of this functionality alone, the user should be able to understand what notes they were playing in their recording and trust the system's analysis of the audio.

Sprint 2

1. **What functionality does the system have at the end of this individual sprint? (2 pts)**

Once Sprint 2 was completed, the system can now analyze and report on uploaded audio. These functionalities are considered to be within the backend of our software. Note pitch analysis is successfully conducted under a multitude of different instrumental presets. The information of the results, along with the various other inputs, is now ready to be sent to the API, which will be implemented in the third sprint.

2. **List the user stories that you successfully implemented during this sprint (1 pt)**
   Stories 9-12

Story ID	User Story
9	Detect Pitch Frequency
10	Identify Letter Notation
11	Evaluate Pitch Accuracy
12	Validate Performance

3. **Did you end up making any changes to any of these user stories? Did you break down any further user stories? Did you identify any new user stories during this sprint and, if so, did you add them to the product backlog or decide to implement them right away? Explain (2 pts)**

Similar to the previous sprint, only minor adjustments were made to the current user stories, and no new stories were created. Due to the more technical nature of this sprint, many adjustments were implemented right away as we learned to use the appropriate tools to build the functionality we were seeking. Foundations have also been made in preparation for Sprint 3.

4. **What are the "lessons learned" at the end of this sprint? What would you do differently next time? Explain (2 pt)**

The primary lesson that was learned was how to determine the “weight” of a user story during this sprint. On our product backlog, we were able to successfully predict the amount of work and priority some of the user stories would have. A lot of the later user stories were heavier in terms of work, and we learned how to successfully split up the work between the 3 sprints, especially after completion of the first sprint. We are confident that we will be able to complete the project by sprint 3.

5. **Provide an updated numbered list of all user stories yet to be implemented; indicate pre- and post- conditions (1 pt)**

Story ID	User Story	Pre-Conditions	Post-Conditions
13	Send Data to Groq API	Performance validation and pitch evaluation results are available in the system. The Groq API connection and AI feedback modules are active.	The validation and results are successfully sent to the Groq API. Additional information is also shared with the API, such as the age group of the user.
14	Generate Feedback Message	The validation and results are received by the Groq API.	Feedback text is generated and stored. Any errors from the API are logged and handled.
15	Display Generated Feedback	Feedback has been generated successfully. Evaluation results are available as well.	Feedback is displayed on the web application, with results for pitch and note accuracy visible. The user can review the results, and the session remains active to be reused if needed.
16	View Feedback Results Page	All the information regarding results, including Pitch Accuracy and Groq API feedback, is loaded in the system.	Information regarding results is still viewable at the end of the session. A new session can be initiated without having to reload the entire application.

6. **If this is not the last sprint:**

Given the current functionality of the system and taking into account the pre- and post-conditions, identify a subset of user stories to be implemented during the next sprint. Be sure that the cumulative size of the selected user stories is about 1/3 of the size of the full backlog. Describe the functionality that your (partially implemented) system will have at the end of this sprint.

By the end of sprint 3, the entire system will be functional from start to finish. The user can submit recorded audio, the system can analyze the pitch, and the information from both the user and the system will be sent to an API to generate feedback based on the user's performance. Different results will be generated based on the information of the user, such as their age group and the instrument used. Every user story should be implemented, allowing for this goal to be achieved.

Sprint 3

1. **What functionality does the system have at the end of this individual sprint? (2 pts)**
   By the end of Sprint 3, the system works all the way through from start to finish. The user records audio, the system analyses the pitch, sends the data to the Groq API, and then displays feedback along with the analysis.
   The feedback is not the same for everyone, it changes based on the age of the user, so younger users get something simple and encouraging while older users get something more detailed. This makes the experience feel personal rather than generic. The user can go through the whole thing in one go record, analyse, and read feedback without leaving the app.

2. **List the user stories that you successfully implemented during this sprint (1 pt)**

   Stories 13-16

Story ID	**User Story**
13	Send Data to Groq API
14	Generate Feedback Message
15	Display Generated Feedback
16	View Feedback Results Page

3. **Did you end up making any changes to any of these user stories? Did you break down any further user stories? Did you identify any new user stories during this sprint and, if so, did you add them to the product backlog or decide to implement them right away? Explain (2 pts)**
   Everything planned was completed. A few things were adjusted along the way:

   3.1. Generate Feedback Message:
   The original plan was more complex, but the Groq API had strict limitations that made it hard to implement fully. The approach was simplified, and even so, the system still gives distinct feedback based on age which was the main goal from the start.

   3.2. Feedback Logic:
   Adjusted so the tone and wording of feedback feels right for different age groups, not just one generic message for everyone.

   3.3. Feedback Page and Results Display:
   Built closely together so moving from the analysis to reading your results feels smooth and connected.

   No new stories were added this sprint.

4. What are the "lessons learned" at the end of this sprint? What would you do differently next time? Explain (2 pt)

The biggest surprise was running into Groq API restrictions. The original feedback logic had to be simplified because of how strict the API rules were. The good news is that even the simpler version still delivers age-based feedback, which was the core idea. With more time in a future sprint, there is room to explore proper workarounds.

The other key lesson was that writing feedback that actually feels helpful to the user is harder than expected. It is not just about getting the right output, it is about saying it in a way that makes sense to whoever is reading it.

4.1. Things to do differently next time:

4.1.1. Research API limitations earlier*:*
A lot of time was spent building the feedback logic before realising the Groq API could not support it the way it was planned. If the limitations were looked into at the start, the approach could have been designed around them from the beginning instead of having to simplify things later on.

4.1.2. Test with a wider range of ages and audio inputs*:*
Most of the testing was done with a limited set of examples, which means there could be cases where the feedback does not quite fit. Testing more variety early on would have caught those gaps sooner and made the feedback more reliable across the board.

4.1.3. Push further on the age-based feedback*:*
The current version works and gives different feedback based on age, but there is a lot more that could be done. With more time and a better understanding of how to work around the API restrictions, the feedback could be much more specific and meaningful for each age group rather than just slightly different wording.

5. Provide an updated numbered list of all user stories yet to be implemented; indicate pre- and post- conditions (1 pt)

No, all user stories from 1 to 16 were completed by the end of this sprint. Nothing was left behind, even with the Groq challenges.

6. Ideas for Future User Stories

Yes, the following user stories could enhance the system further:

6.1. User Story 17: *Customize Feedback Based on Skill Level*

- *Description:* Allow users to select their skill level (beginner, intermediate, advanced) to receive more tailored feedback.

6.2. User Story 18: *Visual Pitch Graph Display*

- *Description:* Display a visual graph showing expected vs detected pitch for better understanding.

6.3. ***User Story 19:*** *Save Feedback History*

- *Description:* Allow users to store and review past performance feedback.

6.4. ***User Story 20:*** *Real-Time Feedback During Recording*

- *Description:* Provide live feedback while the user is recording audio instead of only after completion.